

Nodal Delay in Computer Networks

'nodal delay' from main book (PDF)

Packet loss and Delay

4 components of Nodal Link

- ↳ Transmission ☒
- ↳ Queuing ☒
- ↳ Propagation
- ↳ Processing

How do packet loss and delay occur?

queue full হলে Router new packets রাখতে পারবে না,
drop করে দিবে : congestion loss

arrival rate to link (temporarily) exceeds the
output : packet loss

Four sources of Packet Delay:

$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{trans} :

L: packet length (bits)

R: link transmission rate (bps)

$$\therefore d_{\text{trans}} = L/R$$

d_{prop} :

d: length of physical link

S: propagation speed ($\sim 2 \times 10^8 \text{ m/sec}$)

$$\therefore d_{\text{prop}} = d/s$$

Processing Delay:

Delay to process packet in Router

Switch Interface : 24 (FastEthernet) + 2 (GigabitEthernet)

≈ 26

complex Router

4/8 $\rightarrow$ output

16 $\rightarrow$ input

CRC check

CRC 16

CRC 32

will be

shown

in practical classes

The Internet: A "Service" View

Router 5, Nodes 7
Link 6

Router (n-1)
Link (n)
Node (n+1)

Nodal link * link (n)
nodal delay
 $\left\{ \frac{4}{6} \right\}$
= 24 Delays

If no queues occur,
queueing delay = 0

if in question,
queueing delay is negligible,
 $d_{queue} = 0$

Explains Nodal Delay → Explain all 4 components
E2E Delay

question types

Packet Queueing Delay

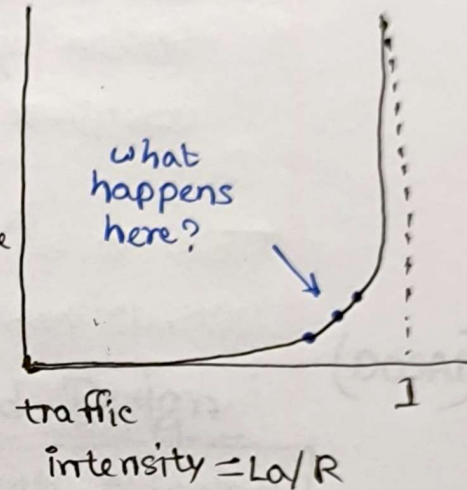
R : link bandwidth (bps)

L : packet length (bps)

α : average packet arrival rate

- * if $L\alpha/R \approx 0$: avg. d_{queue} small
if $L\alpha/R \approx 1$: " " large
if $L\alpha/R > 1$: avg. $d_{\text{queue}} = \infty$

What happens if
traffic intensity is $= 0.5$?
 $L\alpha/R = 0.5$



Throughput = rate at which bits are being sent from sender to receiver.

if $R_s < R_c$

throughput = R_s

{ slide 55 }

Write a short note on throughput
" " " on nodal link

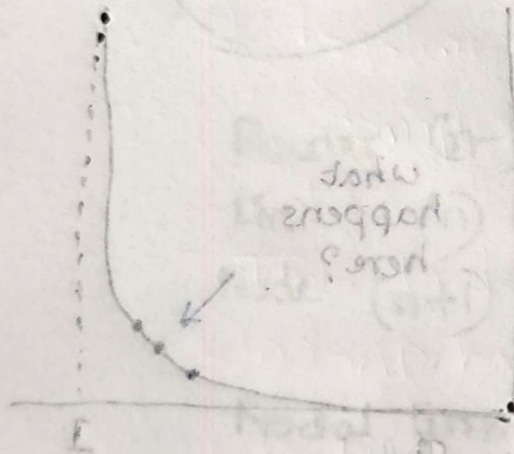
* define Internet

21 Jan Workshop

Registration Fee 150/-

#Internet: A network of networks

Access ISP: general ISP


$$\text{Int}(\mathcal{A}) = \mathcal{A} \cap \mathcal{B}$$

7. $\text{For } R > 1$: avg. degree = ∞
 8. $\text{For } R = 1$: " "
 9. $\text{For } R < 1$: avg. degree small

What happens if traffic intensity is $\rho = 0.5$?

$\rho = 0.5$

weight = 200gms, which is 10% of the total weight.

If no quiescence occurs, reviewer to reviewer

if $R_2 < R_0$

$$R_2 = \text{juggerat}$$